Supervyse OPF Redfish LDAP Setting User Guide

Revision 1.0
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Revision History

Revision Number	Description	Author	Reviewer	Release Date
1.0	Initial release.	Ezra Lin	Victor Wu	12, 2024

1 Introduction

The document's purpose is to help users understand how to add LDAP users into BMC by Redfish.

1.1 Overview

The document's purpose is to help users understand how to add LDAP users into BMC by Redfish.

This document is based on Supervyse OPF Server Setting Guide Chapter 1, please refer to it first.

1.2 Related Information

The following publications and sources of information are referred to by this document or may be useful to you:

- *Supervyse OPF Server Setting Guide, 1.2, Insyde Software, 2024*
- *Supervyse OPF Redfish Api User Guide, 1.38, Insyde Software, 2024*

2 LDAP Server User Groups Tree

Following Supervyse OPF Server Setting Guide Chapter 1, We can create a group "ou=bmc" under "dc=sc2". This group includes "cn=ldap_admin" and "cn=ldap_user" to differentiate between different types of users.

The LDAP admin account is typically "cn=admin,dc=sc2,dc=test", and we can use this account to access the LDAP Server.

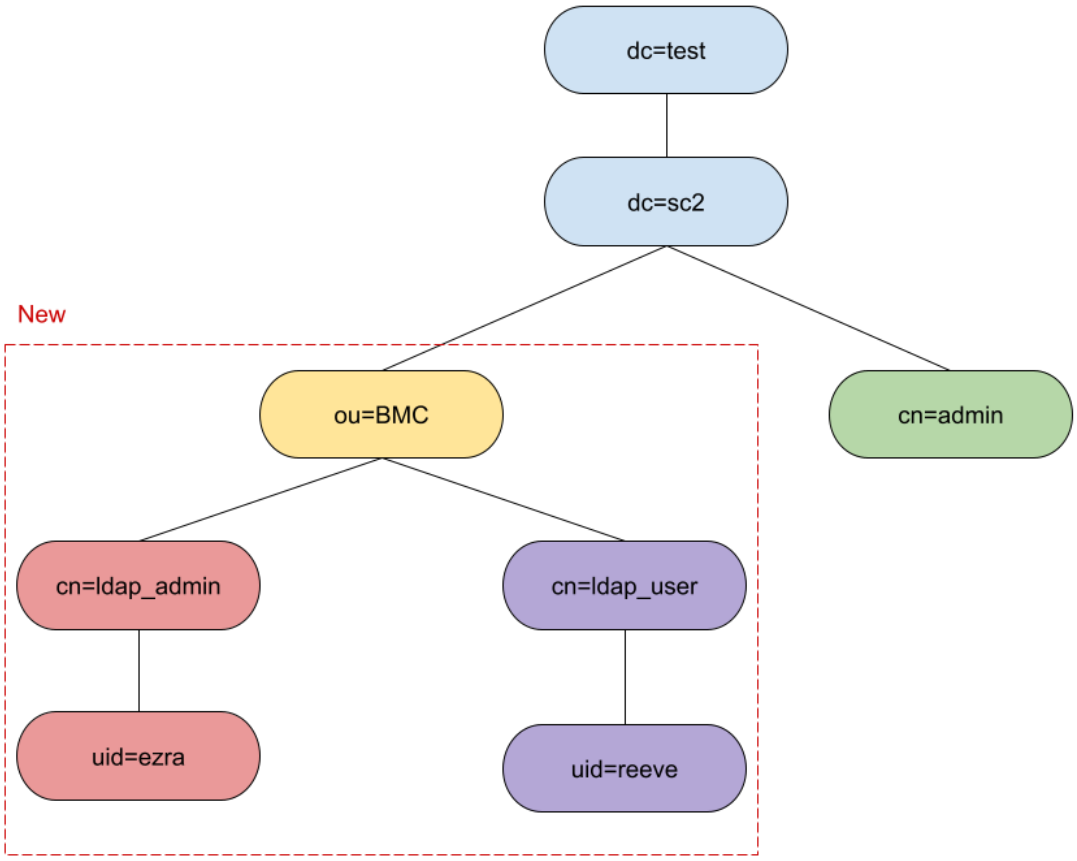

Figure 2-1 LDAP User Group Tree

3 Set LDAP Users Into BMC By Redfish

We can set the LDAP settings on the AccountService resource at the LDAP, RemoteRoleMapping, and the LDAPService objects.

Here is the description of the LDAP, RemoteRoleMapping, Authentication, and LDAPServices objects.

Table 3-1 LDAP, RemoteRoleMapping, Authentication, and LDAPServices Objects Table

Property	Description
LDAP	The first LDAP external account provider this AccountService supports.
→ServiceEnabled	This indicates whether this service is enabled.
→ServiceAddresses	This property contains the addresses of the user account providers this resource references. The format of this field depends on the Type. This value will be displayed if LDAP Service configured.
→Authentication	This property contains the authentication information for the external account provider.
→→AuthenticationType	This property contains the type of authentication used to connect to the external account provider.
→→Username	This property contains the user name for the account service. If AuthenticationType is UsernameAndPassword.
→→Password	This property is used with a PATCH to write the password for the account service. This property is null on a GET. If AuthenticationType is UsernameAndPassword.
→LDAPService	This property contains additional mapping information needed to parse a generic LDAP service.
→→SearchSettings	This property contains the settings needed to search an external LDAP service.
→→→BaseDistinguishedNames	The base distinguished names to use when searching the LDAP service.
→→→UsernameAttribute	The attribute name that contains the Username.
→→→GroupsAttribute	The attribute name that contains the Groups for a user.
→RemoteRoleMapping	This property contains a collection of the mapping rules to convert the external account providers account information to the local Redfish Role.
→→RemoteGroup	This property is the name of the remote group (or in the case of a Redfish Service, remote role) that will be mapped to the local role referenced by this entity. This value will be displayed if LDAP Service configured.
→→LocalRole	The name of the local role in which to map the remote user or group. This value will be displayed if LDAP Service configured.
→→RemoteGroupDomain	The name of the remote group domain, or the remote role in the case of a Redfish service, that maps to the local Redfish role to which this entity links. This value will be displayed if LDAP Service configured.

According to the LDAP server setting, we access the LDAP server to the AccountService resource using the JSON payload below.

```
{
   "LDAP": {
      "ServiceEnabled": true,
      "ServiceAddresses": [
         "ldap://<LDAP Server Address>"
      ],
      "Authentication": {
         "Username": "cn=admin,dc=sc2,dc=test",
         "Password": "<LDAP Server Password>"
      },
   }
}
```

According to the LDAP Server User Groups Tree, We aim to add the users under "ou=BMC,dc=sc2,dc=test" to the BMC and use the LDAP "uid" and "gidNumber" attributes as the username and group on the BMC. At the same time, users under "cn=ldap_admin" will be mapped to the BMC's admin role, and users under "cn=ldap_user" will be mapped to the BMC's user role. Therefore, modifications to the LDAPService->SearchSettings and RemoteRoleMapping objects are required. You can refer to the following example.

Note: The UsernameAttribute and GroupsAttribute can set many values (e.g. uid, cn, gidNumber). However, we suggest setting UsernameAttribute as uid and GroupsAttribute as gidNumber which is a simple and clear method.

```
{
   "LDAP": {
      "LDAPService": {
         "SearchSettings": {
            "BaseDistinguishedNames": [
               "ou=BMC,dc=sc2,dc=test"
            ],
            "UsernameAttribute": "uid",
            "GroupsAttribute": "gidNumber"
         }
      },
      "RemoteRoleMapping": [
         {
            "LocalRole": "Administrator",
            "RemoteGroup": "ldap_admin",
            "RemoteGroupDomain": "ou=BMC,dc=sc2,dc=test"
         },
         {
            "LocalRole": "ReadOnly",
            "RemoteGroup": "ldap_user",
            "RemoteGroupDomain": "ou=BMC,dc=sc2,dc=test"
         }
      ]
   }
}
```

Here is the curl example for adding the LDAP Server to the AccountService.

```
curl -k -u <BMC UserName>:<BMC Password> -X PATCH -H "Content-Type: application/json"
https://<BMC Server Address>/redfish/v1/AccountService -d '{"LDAP": {"ServiceEnabled":
true,"ServiceAddresses": ["ldap://192.168.11.138"],"Authentication": {"Username":
"cn=admin,dc=sc2,dc=test","Password": "admin"},"LDAPService": {"SearchSettings":
{"BaseDistinguishedNames": ["ou=BMC,dc=sc2,dc=test"],"UsernameAttribute":
"uid","GroupsAttribute": "gidNumber"}},"RemoteRoleMapping": [{"LocalRole":
"Administrator","RemoteGroup": "ldap_admin","RemoteGroupDomain": "ou=bmc,dc=sc2,dc=test"},
{"LocalRole": "ReadOnly","RemoteGroup": "ldap_user","RemoteGroupDomain":
"ou=bmc,dc=sc2,dc=test"}]}}'
```

After adding the LDAP Server to the AccountService, you can use ezra and reeve users under LDAP Server to log in to Redfish or Web.